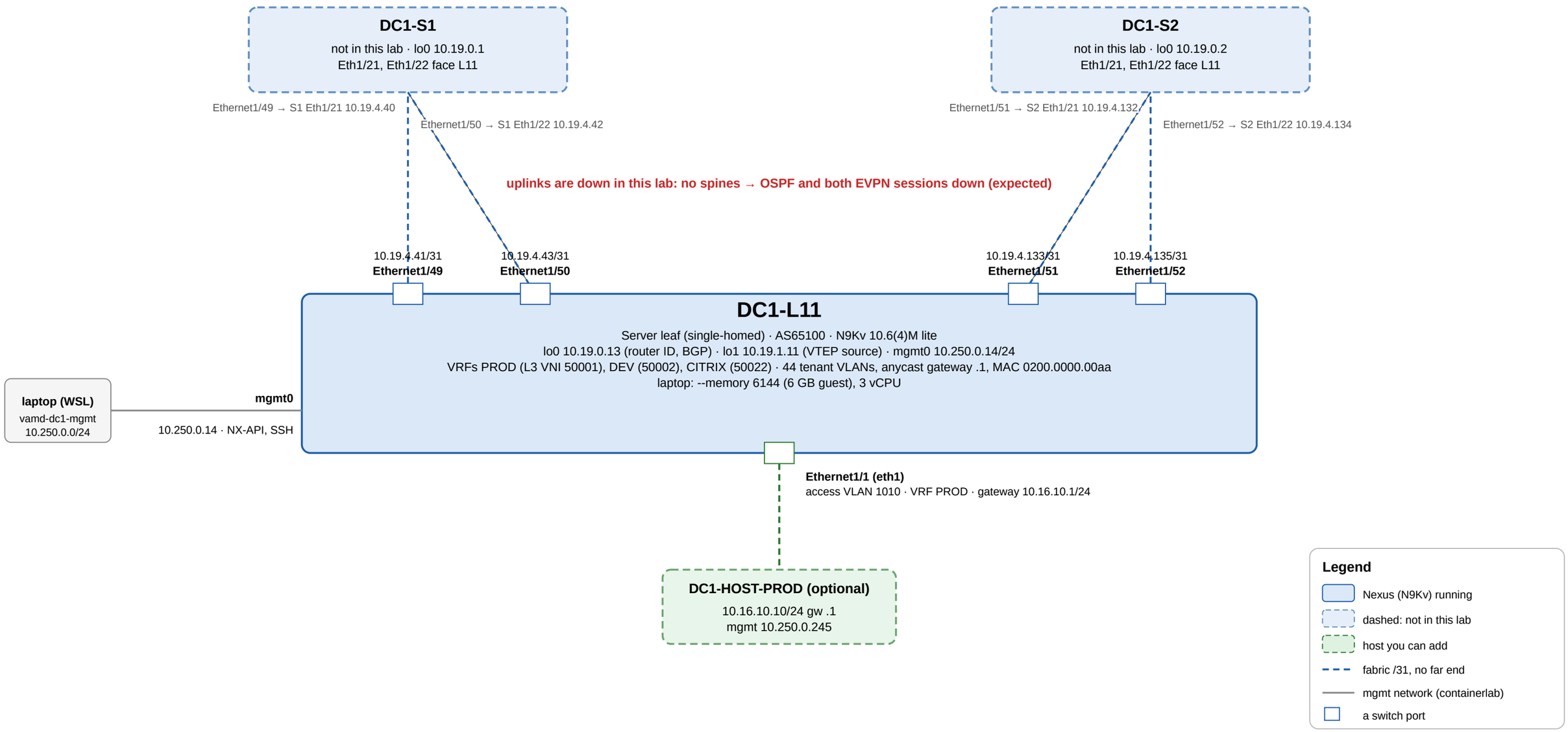
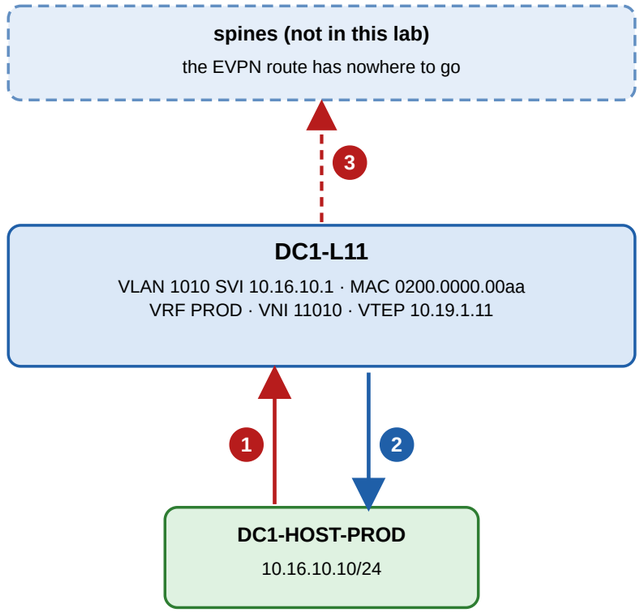
Option 1 · one switch: DC1-L11 on a 12 GB laptop



What this lab is for: loading and checking one rendered NX-OS config on a real NX-OS image (every one of the 106 NX-OS configs was loaded this way), the Day-2 push (`lab/day2/nornir_push.py`), the TCAM carve, and the anycast gateway if you add the PROD host. It cannot show routing between switches: the four uplinks have no far end.

Option 1 · first ARP: the host and its gateway (needs the optional PROD host)



L11 learns the host: MAC table (VLAN 1010, Eth1/1) and ARP (VRF PROD). It builds the EVPN type-2 route, which stays local.

Step	What happens	Look at it
1	The host sends an ARP request for 10.16.10.1 (broadcast in VLAN 1010).	`docker exec -it clab-vamd-dc1-p3-DC1-HOST-PROD ping -c 3 10.16.10.1`
2	L11 answers itself: 10.16.10.1 is at 0200.0000.00aa (anycast gateway). Ran this way in the two-node lab test.	`docker exec -it clab-vamd-dc1-p3-DC1-HOST-PROD ip neigh`
3	L11 learns the host's MAC and IP and makes an EVPN type-2 route (L2 VNI 11010, L3 VNI 50001). With no spines it is not sent anywhere.	`show mac address-table vlan 1010`, `show ip arp vrf PROD`, `show bgp l2vpn evpn`

The rest of the story (the spines reflect the route, other leaves answer ARP from it) needs the fabric: see the DC1 lab path PDF, page 6.

Option 1 · ports, production cables and what stays down

DC1-L11's ports

Port (containerlab: eth + the same number)	Far end	Address / VLAN	State in this lab
Ethernet1/49, 1/50	DC1-S1 Eth1/21, 1/22	10.19.4.41/31, 10.19.4.43/31	down (no far end)
Ethernet1/51, 1/52	DC1-S2 Eth1/21, 1/22	10.19.4.133/31, 10.19.4.135/31	down (no far end)
Ethernet1/1	DC1-HOST-PROD eth1 (optional)	access VLAN 1010	up only if the host is deployed
mgmt0	containerlab mgmt network	10.250.0.14/24	up: NX-API and SSH

The same ports in production (vendor cable schedule)

Cable label	Media	Length	Route
DC1-S1:Eth1/21-DC1-L11:Eth1/49	smf	28 m	DC1-B04-PP42:60 → DC1-C01-PP42:1
DC1-S1:Eth1/22-DC1-L11:Eth1/50	smf	28 m	DC1-B04-PP42:61 → DC1-C01-PP42:2
DC1-S2:Eth1/21-DC1-L11:Eth1/51	smf	31 m	DC1-B08-PP42:63 → DC1-C01-PP42:3
DC1-S2:Eth1/22-DC1-L11:Eth1/52	smf	31 m	DC1-B08-PP42:64 → DC1-C01-PP42:4

What is configured but cannot come up

Item	Configured	In this lab
OSPF UNDERLAY	area 0 on the four uplinks and loopbacks	no neighbours
iBGP EVPN	neighbours 10.19.0.1 and 10.19.0.2 (the spines), AS 65100, BFD multihop	Idle / Active, BFD down
NVE	nve1, source loopback1, 44 L2 VNIs + 3 L3 VNIs, ingress replication BGP	up, no peers (`show nve peers` empty)
Default routes in PROD / DEV / CITRIX	learned from the service leaves through EVPN	absent

Option 1 · DC1-L11's overlay: VRFs, VLANs, VNIs

VRF	L3 VNI	L3VNI VLAN	Tenant VLANs	L2 VNIs	Subnets (anycast gateway .1)
PROD	50001	2501	1010–1017, 1040–1051	11010–11017, 11040–11051	10.16.10–17.0/24, 10.16.40–51.0/24
DEV	50002	2502	1110–1118, 1140–1150	11110–11118, 11140–11150	10.16.74–82.0/24, 10.16.104–114.0/24
CITRIX	50022	2522	1500–1501	11500–11501	10.16.144–145.0/24

44 tenant VLANs on DC1-L11, each with an SVI at .1 (MAC 0200.0000.00aa), `fabric forwarding mode anycast-gateway` and ARP suppression on the NVE. L2 VNI = 10000 + VLAN. The PROD host uses VLAN 1010. Check on the switch: `show nve vni`, `show vrf`, `show ip interface brief vrf PROD`.

Option 1 · run card (12 GB laptop)

What you need

Item	Value
Memory	About 7 GB free in WSL with Infrahub stopped (guest 6 GB with `--memory 6144`, plus a 512 MB reserve)
Software	containerlab 0.79, Docker, `vrnetlab/cisco_n9kv:10.6.4.M-lite`
Time	About 15 minutes from `deploy` to a configured switch
Topology file used	`lab/build/DC1/topology.override.yml` (written because of `--memory`)

Bring it up

```
# fetch while Infrahub is up, then free the memory
.venv/bin/python lab/labctl.py fetch DC1 --compare-baselines
docker compose stop
# the switch alone (add ,DC1-HOST-PROD for the gateway test)
.venv/bin/python lab/labctl.py deploy DC1 --node-filter DC1-L11 --memory 6144 --reserve 512
.venv/bin/python lab/labctl.py push DC1 --nodes DC1-L11
.venv/bin/python lab/labctl.py carve DC1 --nodes DC1-L11
```

Take it down

```
.venv/bin/python lab/labctl.py destroy DC1
docker compose start
```

Log in

```
ssh ciscoadmin@10.250.0.14     # password from ztp/make-secret.sh
```

Stop and start

`containerlab stop -t lab/build/DC1/topology.override.yml --node DC1-L11` works; the matching `start` did not bring the switch back in 40 minutes (laptop test, 19 Oct). To get the switch back, run destroy, then deploy, push and carve.

What ran this way (laptop)

Test	Result
All 106 NX-OS configs loaded on one N9Kv (acceptance test)	**pass**
Two-node lab: L11 + PROD host pings its anycast gateway	**pass**
Day-2 push: config replace, then `--check` "Rollback Patch is Empty"	**pass (10 Oct, 19 Oct)**
Per-node stop/start of the Nexus	**start failed (19 Oct)**

Known issue: `labctl push` needs netmiko for its SSH fallback; the laptop's `.venv` does not have it (the carve re-push covers it).